

CTU-Chat: A Hybrid Retrieval-Augmented Generation System for University Student Services

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Abstract. University administrative question answering demands more than semantic similarity matching: answers must respect document validity, organizational authority, temporal scope, and exact procedural or numerical rules. This paper presents CTU-CHAT, a bilingual Vietnamese–English framework that integrates metadata-filtered BM25 and dense retrieval, a provenance-preserving knowledge graph, cross-encoder evidence reranking, and a specialist-agent workflow. PostgreSQL serves as the authoritative data store, while Qdrant and Neo4j function as rebuildable retrieval projections; every candidate is revalidated against publication, approval, audience, category, and effective-date constraints prior to generation. The framework is evaluated through retrieval and end-to-end QA studies using 150 human-reviewed questions spanning nine administrative categories. The full retrieval configuration achieves H@1 of 0.8667, perfect H@3 and Recall@5, and MRR@10 of 0.9222. In the end-to-end evaluation, the proposed system attains Context Recall of 0.745 and Context Precision of 0.749, while removing governance produces the lowest Answer Correctness score. This work provides a reproducible design for evidence-grounded administrative assistants in low-resource bilingual settings.

Keywords: Retrieval-augmented generation · GraphRAG · Agentic AI · Evidence provenance · Hybrid retrieval · Bilingual question answering · University administration · Vietnamese NLP

1 Introduction

Providing accurate and timely administrative support is an essential component of modern university student services. Students frequently seek information about tuition fees, scholarships, student loans, academic regulations, curricula, and administrative procedures. However, these answers are distributed across versioned regulations, official announcements, fee tables, and curriculum documents, making reliable information retrieval a challenging task. Incorrect or outdated responses may directly affect students’ academic and financial decisions.

Recent advances in Retrieval-Augmented Generation (RAG) [4] have significantly improved question answering by grounding large language models on external knowledge. Dense retrieval effectively captures semantic similarity, while

lexical retrieval remains valuable for matching exact identifiers such as course codes, decision numbers, and Vietnamese administrative terminology. Knowledge graphs and agent-based orchestration further extend RAG by supporting relational reasoning and specialized tool invocation.

Despite these advances, existing RAG systems remain insufficient for university administrative services. First, semantic retrieval alone often fails to retrieve exact policy identifiers and structured tabular information. Second, most systems do not preserve document governance, including version validity, publication status, and effective dates. Third, they lack deterministic mechanisms for numerical calculations such as tuition reduction and scholarship eligibility, leading to fluent but potentially incorrect responses. These limitations reveal a research gap in building evidence-grounded and governance-aware RAG systems for higher education.

Research Question. *How can a hybrid retrieval framework combining lexical search, dense retrieval, knowledge graphs, and deterministic tools improve the reliability and factual accuracy of university student-service question answering?*

To address this question, this paper proposes **CTU-Chat**, a hybrid Retrieval-Augmented Generation framework for Can Tho University. The proposed system integrates Vietnamese BM25, dense vector retrieval, Reciprocal Rank Fusion (RRF), metadata-governed evidence filtering, graph-based relational retrieval, and intent-aware multi-agent orchestration with typed calculation tools. This architecture preserves evidence provenance while providing accurate, transparent, and reproducible answers for administrative and academic queries.

The main contributions of this work are as follows:

- A governance-aware hybrid retrieval architecture that combines lexical, semantic, and graph-based evidence under temporal and metadata constraints.
- An intent-aware multi-agent framework integrating Neo4j graph retrieval with deterministic academic and financial calculation tools.
- A comprehensive experimental evaluation on a human-verified university student-service benchmark to assess retrieval quality, answer correctness, and system reliability.

2 Related Work

This section reviews representative studies on Retrieval-Augmented Generation, hybrid retrieval, GraphRAG, agentic AI, and educational question answering. Rather than summarizing previous work, we analyze their limitations and identify the research gap addressed by CTU-Chat.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG), first introduced by Lewis et al. [4], combines external document retrieval with large language models to generate

evidence-grounded responses. By conditioning generation on retrieved passages, RAG significantly improves factual accuracy and reduces hallucinations in open-domain question answering.

However, conventional RAG assumes a static and homogeneous document collection in which all retrieved documents are treated as equally reliable. Such an assumption is unsuitable for university administrative services because regulations are continuously revised and possess different publication states and periods of validity. Consequently, existing RAG systems rarely preserve document provenance or distinguish obsolete documents from currently effective regulations, limiting their reliability in governance-oriented applications.

2.2 Lexical and Dense Retrieval for Bilingual Text

Lexical retrieval and dense retrieval represent two complementary paradigms for information retrieval. BM25 [6] remains highly effective for matching exact policy terminology, course identifiers, and administrative codes, while multilingual dense embeddings [1] provide robust semantic matching across Vietnamese and English paraphrases.

The BM25 ranking function is defined as:

$$s_{\text{BM25}}(q, d) = \sum_{w \in q} \text{IDF}(w) \frac{\text{tf}(w, d)(k_1 + 1)}{\text{tf}(w, d) + k_1 \left(1 - b + b \frac{|d|}{|d|}\right)}. \quad (1)$$

Dense retrieval computes semantic similarity using cosine similarity between query and document embeddings:

$$s_{\text{dense}}(q, d) = \frac{\mathbf{h}_q^\top \mathbf{h}_d}{\|\mathbf{h}_q\|_2 \|\mathbf{h}_d\|_2}. \quad (2)$$

To leverage the strengths of both paradigms, hybrid retrieval commonly employs Reciprocal Rank Fusion (RRF) [2] to merge lexical and semantic rankings without score normalization.

Despite their complementary advantages, existing hybrid retrieval methods primarily optimize relevance while overlooking governance metadata. Highly ranked passages may therefore originate from expired or unauthorized documents, and Vietnamese-specific challenges such as word segmentation and compound administrative terminology remain insufficiently addressed by general multilingual retrieval models.

2.3 Knowledge Graphs and GraphRAG

Knowledge graphs extend retrieval by explicitly modeling entities and their relationships, enabling multi-hop reasoning across organizational structures, curricula, prerequisites, and administrative procedures. GraphRAG [3] integrates graph traversal with retrieval-augmented generation to discover evidence beyond isolated text chunks.

Nevertheless, most GraphRAG frameworks directly project LLM-extracted facts into the graph without validating their correctness. Unreviewed graph edges may propagate inaccurate relationships during retrieval, while graph-expanded candidates frequently bypass the governance constraints enforced in the authoritative document repository. These issues reduce the trustworthiness of graph reasoning in highly regulated domains such as university administration.

2.4 Agentic RAG and Multi-Agent Systems

Agentic RAG extends traditional retrieval pipelines through autonomous planning, tool invocation, and iterative evidence acquisition [7]. Supervisor–Worker orchestration frameworks, such as LangGraph, further improve scalability by assigning specialized responsibilities to domain-specific agents with dedicated tools and contextual memory.

Although these architectures provide greater flexibility than conventional RAG, increased autonomy also introduces new sources of uncertainty. Routing errors, inappropriate tool selection, memory leakage across sessions, and non-deterministic reasoning negatively affect reproducibility. Moreover, existing studies rarely evaluate whether an agent should abstain or retry retrieval when supporting evidence is insufficient, leaving reliability largely unexplored.

2.5 Administrative and Educational Question Answering

Educational question answering systems have been widely applied to university student services, including academic advising, curriculum consultation, and administrative assistance [5]. These systems demonstrate the practicality of conversational AI for accessing institutional knowledge and supporting student services.

However, most existing educational QA systems remain document-centric rather than governance-centric. They seldom support bilingual Vietnamese–English administrative language, preserve citation provenance, or integrate deterministic calculation tools for tuition and GPA policies. In addition, multi-agent routing and evidence-aware response generation are rarely evaluated, indicating that reliable university governance remains an open research problem.

2.6 Research Gap and Problem Formulation

The literature review reveals that no existing approach simultaneously addresses multilingual retrieval, document governance, provenance preservation, temporal validity, and deterministic policy computation within a unified university student-service framework. This study therefore formulates the administrative QA problem as follows.

Let a user query be defined as $q = (x, \ell, \mathcal{C}, \mathcal{A}, t)$, where x is the question text, $\ell \in \{vi, en\}$ is the language, $\mathcal{C}$ denotes category constraints, $\mathcal{A}$ represents an optional audience constraint, and t is the reference time for answer validity.

The objective is to generate an answer y together with a set of supporting citations $Z \subseteq E$ such that every cited evidence is relevant, originates from an approved and effective document version, satisfies governance constraints, and fully supports the generated claims. When sufficient evidence is unavailable, the system should abstain or explicitly express uncertainty rather than generating unsupported information.

The governance constraint is formalized by the predicate:

$$G(e, q) = I_{\text{approved}}(e) \cdot I_{\text{published}}(e) \cdot I_{\text{category}}(e, \mathcal{C}) \cdot I_{\text{audience}}(e, \mathcal{A}) \cdot I_{\text{effective}}(e, t). \quad (3)$$

Only evidence satisfying $G(e, q) = 1$ is allowed to participate in answer generation. This formulation establishes the theoretical foundation for the governance-aware hybrid retrieval architecture proposed in the following section.

3 Proposed System Architecture

4 Methodology

This section describes the core methodologies implemented in CTU-Chat, including hybrid retrieval, cross-encoder evidence reranking, governed retrieval lanes, graph-based candidate expansion, and domain-specific deterministic calculation tools.

4.1 Hybrid Retrieval with Reciprocal Rank Fusion

To improve retrieval robustness, CTU-Chat combines BM25 lexical retrieval and dense semantic retrieval. Since these retrievers produce heterogeneous score distributions, their scores are not directly summed. Instead, retrieved candidates are merged, deduplicated by document identifier, and ranked using Reciprocal Rank Fusion (RRF). Graph-derived candidates are added later by the orchestration layer and are reranked with the fused text candidates.

For the retrieval source set $\mathcal{S} = \{\text{BM25}, \text{Dense}\}$, the RRF score of a candidate document d is computed as

$$s_{\text{RRF}}(d) = \sum_{s \in \mathcal{S}_d} \frac{1}{k + r_s(d)}, \quad k = 60, \quad (4)$$

where $r_s(d)$ denotes the ranking position of document d returned by retrieval source s , and $\mathcal{S}_d$ contains only the retrievers that successfully retrieve d . The constant $k = 60$ is fixed across all experiments to ensure reproducibility.

The query-intent classifier selects metadata-constrained retrieval lanes for tuition, scholarships, loans, social support, academic programs, and academic regulations. For supported academic-program queries, Neo4j entity lookup adds program metadata to the candidate set. The graph is treated as an expansion mechanism rather than an authority source.

4.2 Cross-Encoder Evidence Reranking

Following RRF fusion and governed-lane expansion, CTU-Chat uses BAAI/`bge-reranker-v2-m3` to score each query–document pair. The implementation truncates the text used for scoring to 1,000 characters while retaining the full document for answer generation. Documents are ranked by cross-encoder relevance; when scores differ by at most 0.05, newer timestamped documents are preferred. This neural reranker is not the hand-weighted deterministic formula described in earlier drafts.

4.3 Deterministic Academic and Financial Tools

Certain university services require exact numerical computation rather than language-model reasoning. CTU-Chat therefore integrates typed deterministic tools for course grading, GPA calculation, and tuition fee estimation, ensuring reproducible results that are directly traceable to institutional policies.

Course Grade Calculation Given assessment scores x_i and weights w_i satisfying $\sum_i w_i = 100$, the final 10-point score is calculated as

$$S_{10} = \text{Round}_{0.1} \left(\sum_i \text{Round}_{0.1}(x_i) \frac{w_i}{100} \right). \quad (5)$$

The resulting score is converted into the official CTU letter-grade scale (A, B+, B, C+, C, D+, D, and F).

GPA Calculation For eligible courses $\mathcal{I}$ with credits c_i and grade points g_i , GPA is computed as

$$\text{GPA} = \text{Round}_{0.01} \left(\frac{\sum_{i \in \mathcal{I}} c_i g_i}{\sum_{i \in \mathcal{I}} c_i} \right). \quad (6)$$

Courses with statuses Missing (M), Incomplete (I), and Withdrawn (W) are excluded according to the university grading policy.

Tuition Fee Calculation For each registered course i , the payable tuition fee is determined from the credit load, tuition rate, exemption basis, and reduction percentage:

$$\text{Gross}_i = c_i r_i m_i, \quad (7)$$

$$\text{Reduction}_i = c_i b_i \frac{p_i}{100}, \quad (8)$$

$$\text{Payable}_i = \text{Round}_{1 \text{ VND}} (\max(0, \text{Gross}_i - \text{Reduction}_i)). \quad (9)$$

All computations use decimal arithmetic, and every tool output preserves the corresponding evidence identifier and policy reference to guarantee auditability.

5 Experimental Results

This section presents the experimental dataset, implementation environment, evaluation metrics, experimental scenarios, and discussion of the proposed CTU-Chat framework.

5.1 Experimental Dataset

A benchmark dataset was constructed to evaluate the proposed CTU-Chat framework on university student-service question answering. The dataset was developed from official Can Tho University documents, including academic regulations, training programs, tuition policies, scholarships, student loan policies, social support, dormitory regulations, and administrative procedures.

Instead of manually composing every question, candidate questions were first generated using Claude based on the collected documents. The generated dataset then underwent a two-stage validation process. First, a validation notebook was used to verify the availability of supporting documents and evidence references. Second, all questions were manually reviewed to remove duplicated or ambiguous samples and to ensure that each question had a complete ground-truth answer. A final validation step was performed before constructing the benchmark.

The final benchmark contains 150 representative questions selected according to three criteria: (1) coverage of all major administrative categories, (2) high-frequency real-world student inquiries, and (3) complete ground-truth evidence for every sample.

Table 1. Distribution of the benchmark dataset.

Category	Questions
Actual Tuition	60
Academic Rules	18
Scholarship	14
Student Loan	12
Other	12
Social Support	11
Academic Program	9
Exemption Policy	9
Exemption Basis	5
Total	150

5.2 Experimental Tools

The proposed system was implemented using a hybrid retrieval architecture consisting of vector, graph, and relational databases. Dense embeddings were

generated using the Vietnamese bi-encoder used by the deployed index, and a BGE reranker was used in the upgraded retrieval pipeline. LangGraph was used for multi-agent orchestration. Retrieval metrics were computed from gold document identifiers; answer relevancy and correctness were scored by a Gemini-based LLM judge.

Table 2. Experimental implementation environment.

Component	Technology
Programming language	Python 3.12
Vector database	Qdrant
Graph database	Neo4j
Metadata database	PostgreSQL
Embedding model	vietnamese-bi-encoder
Cross-encoder reranker	bge-reranker-v2-m3
Generation and judge LLM	Gemini 2.5 Flash Lite
Agent framework	LangGraph
Evaluation	Gold-source metrics + LLM judge

5.3 Evaluation Metrics

Two groups of evaluation metrics were adopted in this study.

Retrieval metrics evaluate the quality of evidence retrieval, including Hit@1 (H@1), Hit@3 (H@3), Precision@5 (P@5), Recall@5, and Mean Reciprocal Rank (MRR@10).

End-to-end QA metrics follow the terminology of Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). In the reported run, CR and CP were computed deterministically from gold document identifiers, whereas AR and AC were assigned by a Gemini 2.5 Flash Lite judge. These are therefore RAGAS-style metrics, not scores produced by the RAGAS software package. All metrics range from 0 to 1, where higher values indicate better performance.

5.4 Experimental Scenarios

Two groups of experiments were conducted using the same 150-question benchmark. The first evaluates the retrieval module (E1–E5), while the second evaluates the complete question answering pipeline (T1–T7).

Scenario A: Retrieval Performance Five cumulative retrieval configurations were evaluated. E1 uses BM25, E2 uses dense retrieval, and E3 combines both retrievers through RRF. E4 adds the **bge-reranker-v2-m3** cross-encoder, while

E5 represents the complete system with governed retrieval lanes, graph-based candidate expansion, and cross-encoder reranking.

Table 3 shows a clear improvement as the reranker and full system components are introduced. E4 raises H@1 from 0.6000 to 0.8000 and MRR@10 from 0.7006 to 0.8556 relative to E3. E5 achieves the strongest overall retrieval result, including perfect H@3 and Recall@5 and an MRR@10 of 0.9222.

Table 3. Retrieval evaluation results of E1–E5 on 150 questions.

Config.	H@1	H@3	P@5	R@5	MRR
E1	0.7333	0.8667	0.2133	0.9000	0.8133
E2	0.5333	0.6000	0.1600	0.6667	0.5833
E3	0.6000	0.8000	0.2000	0.8333	0.7006
E4	0.8000	0.9333	0.2267	0.9333	0.8556
E5	0.8667	1.0000	0.2400	1.0000	0.9222

Scenario B: End-to-End Question Answering The second experiment evaluates seven QA configurations (T1–T7) using the metric procedure described above. The logged run contains 150 cases for each configuration, for a total of 1,050 generated answers and 1,050 judge records.

Table 4 shows that T4 achieves strong context quality, with Context Recall of 0.745 and Context Precision of 0.749. In contrast, removing the governance filter in T7 reduces Answer Correctness to 0.398, the lowest score in the experiment. T5 and T6 remain competitive on answer correctness, indicating that high-quality retrieval and answer generation must be optimized jointly.

Table 4. End-to-end QA results of T1–T7 on 150 questions.

Config.	AR	CR	CP	AC
T1	0.561	0.745	0.684	0.518
T2	0.455	0.533	0.421	0.395
T3	0.578	0.682	0.550	0.506
T4	0.551	0.745	0.749	0.500
T5	0.575	0.682	0.550	0.525
T6	0.561	0.745	0.750	0.508
T7	0.479	0.533	0.421	0.398

5.5 Discussion

The retrieval results demonstrate the complementary roles of lexical matching, semantic retrieval, and learned reranking. BM25 remains effective for exact co-

hort identifiers and administrative terminology, whereas the cross-encoder substantially improves the ranking of hybrid candidates. Governed lanes and graph expansion further improve evidence coverage, allowing E5 to retrieve all required sources within the top five for this benchmark.

The end-to-end results reinforce the importance of governed retrieval. T4 maintains high Context Recall and Context Precision, while T7 records the lowest Answer Correctness. At the same time, the competitive AC values of T5 and T6 show that context quality alone does not guarantee the best generated answer; generation prompts and answer calibration remain important optimization targets.

The benchmark is drawn from a single institution and contains a large tuition subset. Future work will extend the evaluation to balanced bilingual queries, additional institutions, and repeated or human-assisted answer assessment.

6 Conclusion

This paper presented **CTU-Chat**, a hybrid Retrieval-Augmented Generation (RAG) system for university student services that integrates lexical, dense, and graph retrieval with governed document management for bilingual Vietnamese–English administrative question answering.

The proposed framework addresses five research gaps identified in existing educational RAG systems, namely the absence of document governance, limited integration of lexical, semantic, and graph retrieval, insufficient provenance preservation, lack of evidence-aware retrieval, and the reliance on LLM-based numerical reasoning for academic and financial services.

On the 150-question retrieval benchmark, the complete E5 configuration achieved H@1 of 0.8667, perfect H@3 and Recall@5, and MRR@10 of 0.9222. Compared with the hybrid E3 baseline, the cross-encoder in E4 substantially improved ranking quality, while governed lanes and graph-based expansion in E5 provided the strongest overall retrieval performance.

In the end-to-end evaluation, T4 achieved Answer Relevancy of 0.551, Context Recall of 0.745, Context Precision of 0.749, and Answer Correctness of 0.500. The T7 configuration without governance produced the lowest Answer Correctness score of 0.398, highlighting the value of cohort-aware and domain-aware evidence selection for reliable administrative question answering.

Future work will expand the benchmark to additional institutions, add balanced Vietnamese–English test cases, improve answer generation from high-quality contexts, and enhance graph construction for more complex multi-hop reasoning.

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